Scenario Based Assignment

Question 1: Campus Delivery Robot

a) Name the search strategy and justify your answer

The search strategy used by the delivery robot is Greedy Best-First Search.

Greedy Best-First Search selects the next location based only on its heuristic value \(h(n)\), which represents the straight-line distance from the current location to the Auditorium. It does not consider the distance already travelled.

The formula is:

f(n)=h(n)

The robot always selects the unvisited neighbouring location with the lowest straight-line distance to the Auditorium.

b) Trace the robot's decisions in a table

Current location	Candidate location	Straight-line distance \(h(n)\)
Gate	Library	330 m
Gate	Canteen	240 m
Canteen	Hostel	160 m
Canteen	Admin	110 m
Admin	Auditorium	0 m

The robot makes the following decisions:

1. It starts at Gate. Between Library (330 m) and Canteen (240 m), it chooses Canteen.
2. At Canteen, it compares the unvisited neighbours Hostel (160 m) and Admin (110 m). It chooses Admin.
3. From Admin, it reaches the Auditorium, whose heuristic value is 0.

c) State the route and total distance. Is it guaranteed to be the shortest?

The route selected is:

Gate → Canteen → Admin → Auditorium

Total distance:

$$200 + 260 + 120 = \boxed{580\ \text{m}}$$

This route is not guaranteed to be the shortest because Greedy Best-First Search considers only the straight-line heuristic and ignores the distance travelled.

For comparison, another valid route is:

Gate → Canteen → Hostel → Auditorium

$$200 + 100 + 180 = \boxed{480\ \text{m}}$$

Therefore, the greedy route of 580 m is not the shortest route.

Question 2: Campus Delivery Robot Using A* Search

a) Identify the new strategy and explain why it can guarantee the shortest route

The new search strategy is A Search*.

A* Search considers both the distance already travelled and the estimated straight-line distance remaining to the destination. It uses the formula:

f(n)=g(n)+h(n)

Where:

- g(n) = actual distance travelled from the starting point.
- h(n)= estimated straight-line distance to the Auditorium.
- f(n)= estimated total route cost.

A* can find the shortest route when the heuristic is admissible, the edge costs are non-negative, and the algorithm correctly selects the lowest-cost candidate from its frontier. Here, the straight-line heuristic does not overestimate the actual remaining distance.

b) Trace the new rule in a table

The robot starts at Gate.

Step	Candidate	Distance travelled g(n)	Heuristic h(n)	Total f(n)
1	Library	120	330	450
1	Canteen	200	240	440
2	Library	120	330	450
2	Hostel	300	160	460
2	Admin	460	110	570
3	Lab	270	280	550
4	Auditorium via Hostel	480	0	480

Step-by-step explanation:

1. From Gate, A* compares Library (450) and Canteen (440). It selects Canteen.
2. After expanding Canteen, Library still has the lowest frontier value of 450, so A* expands Library.
3. From Library, it discovers Lab with a value of 550. The longer route to Hostel through Library is not preferred.
4. A* then selects Hostel, with a value of 460.
5. From Hostel, it reaches Auditorium at a total route cost of 480 m. This is the shortest route.

c) Check the heuristic values and compare the routes

The straight-line estimates are compared with the actual shortest remaining road distances.

Location	Heuristic h(n)	Actual shortest remaining distance	Safe?
Gate	400 m	480 m	Yes
Library	330 m	480 m	Yes
Canteen	240 m	280 m	Yes
Hostel	160 m	180 m	Yes

The heuristic is safe because it does not overestimate the actual shortest remaining distance for these locations.

- Question 1 route: Gate → Canteen → Admin → Auditorium = 580 m.
- Question 2 route: Gate → Canteen → Hostel → Auditorium = 480 m.

Therefore, A* finds a route that is 100 m shorter than the greedy route.

Question 3: Workshop Scheduling Using CSP

a) Identify the problem and formulate it

This is a Constraint Satisfaction Problem (CSP) because each workshop must be assigned a time slot while satisfying the given constraints.

Variables: W1, W2, W3, W4, W5.

Domains:

- W1: Morning, Afternoon, Evening
- W2: Morning, Afternoon, Evening
- W3: Afternoon, Evening
- W4: Morning, Afternoon
- W5: Morning, Afternoon, Evening

Constraints:

1. W1 and W2 cannot have the same slot.

2. W1 and W3 cannot have the same slot.
3. W2 and W3 cannot have the same slot.
4. W2 and W4 cannot have the same slot.
5. W3 and W5 cannot have the same slot.
6. W4 and W5 cannot have the same slot.
7. W3 cannot be scheduled in the Morning.
8. W4 cannot be scheduled in the Evening.

b) Build a valid timetable using the given selection rules

The workshop with the fewest remaining legal slots is selected first. Ties are broken by the number of unscheduled conflicting workshops and then by workshop number. Slots are tried in Morning, Afternoon, Evening order.

Step	Selected workshop	Assigned slot	Remaining domains of unscheduled workshops
1	W3	Afternoon	W1: M, E; W2: M, E; W4: M, A; W5: M, E
2	W2	Morning	W1: E; W4: A; W5: M, E
3	W1	Evening	W4: A; W5: M, E
4	W4	Afternoon	W5: M
5	W5	Morning	None

Here, M = Morning, A = Afternoon and E = Evening.

Final timetable:

Workshop	Assigned time slot
W1	Evening
W2	Morning

W3	Afternoon
W4	Afternoon
W5	Morning

This timetable satisfies all the constraints. No conflicting workshops share a slot, W3 is not in the Morning, and W4 is not in the Evening.

c) Explain the benefit of choosing the workshop with the fewest remaining slots

The Minimum Remaining Values (MRV) heuristic selects the workshop with the fewest legal time slots first. This helps identify difficult assignments early and reduces unnecessary searching.

If any workshop has no legal slot remaining, the current assignment cannot produce a valid timetable. The algorithm must backtrack by undoing an earlier assignment and trying another available slot.

Question 4: Minimax Algorithm for a Two-Player Game

a) Identify the technique and state its assumption

The technique is the Minimax algorithm.

Minimax is used in two-player, turn-based games to select the best move by considering the possible responses of the opponent.

It assumes that both players play optimally:

- Player X tries to maximize the score.
- Player O tries to minimize Player X's score.
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b) Calculate the value of every node

The final scores from left to right are:

4, 7 | 8, 2 | 3, 5 | 6, 9

Step 1: Evaluate Player X's final moves.

Player X selects the maximum score from each pair.

Final scores	X's selected value
4, 7	7
8, 2	8
3, 5	5
6, 9	9

Step 2: Evaluate Player O's replies.

Player O selects the minimum value from each pair of X's outcomes.

X's possible outcomes	O's selected value
7, 8	7
5, 9	5

Step 3: Evaluate the starting position.

Player X selects the maximum of the two values:

$$\max(7,5) = \boxed{7}$$

Therefore, the value of the starting position is 7.

c) State X's best first move and the resulting sequence

- Best first move: X's first move.
- Guaranteed score: 7.
- Optimal sequence: X selects the first move, O selects the reply with outcomes 4 and 7, and X chooses the final score of 7.

Player X can guarantee a score of 7 when both players play optimally.

Question 5: Alpha-Beta Pruning

a) Evaluate the moves from left to right and mark skipped positions

Alpha-Beta Pruning is an improvement of the Minimax algorithm. It avoids evaluating positions that cannot change the final decision.

The engine is the MAX player, and the opponent is the MIN player.

The given scores are:

- M1: 6, 9, 7
- M2: 4, 8, 10
- M3: 7, 5, 12

Step 1: Evaluate M1

The opponent selects the minimum score.

$$\min(6,9,7) = 6$$

The best value found so far for the engine is 6, so \(\alpha=6\).

Step 2: Evaluate M2

The first score is 4. Since the opponent can force a value of 4, which is less than the engine's current best value of 6, the remaining scores can be skipped.

- Evaluated: 4
- Pruned: 8, 10
- M2 value: 4

Step 3: Evaluate M3

The first score is 7, and the second score is 5. The opponent can already force a value of 5, which is less than the engine's current best value of 6. Therefore, the last score can be skipped.

- Evaluated: 7, 5

- Pruned: 12
- M3 value: 5

Move	Scores examined	Scores skipped	Final value
M1	6, 9, 7	None	6
M2	4	8, 10	4
M3	7, 5	12	5

b) State the number of positions evaluated and the chosen move

Total possible positions = 9.

Positions actually evaluated:

$$3 + 1 + 2 = \boxed{6}$$

Positions skipped:

$$9 - 6 = \boxed{3}$$

The engine selects M1, with a value of 6, because it has the highest guaranteed score.

Without pruning, Minimax would also select M1 with a value of 6. Thus, Alpha-Beta Pruning reaches the same decision while examining fewer positions.

c) Explain how move order affects pruning and why an evaluation function is needed

The order in which moves are examined affects how many positions can be skipped. If promising moves are evaluated first, better alpha and beta bounds are established earlier, allowing more branches to be pruned.

An evaluation function is needed when the complete game tree cannot be searched. It estimates the usefulness of a position and assigns a numerical score. Higher scores represent better positions for the engine in this assignment.